Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The triangle formed by the tangent to the curve $f(x) = x^2 + bx - b$ at the point $(1, 1)$ and the coordinate axes, lies in the first quadrant. If its area is 2, then the value of b is

(A) -1

(B) 3

(C) -3

(D) 1

Q.2

The area enclosed by the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$, $\frac{x-7^2}{16} + \frac{y^2}{4} = 1$ and $y = 2$ is

(A) $\frac{28-\pi}{2}$ sq. units

(B) $\frac{28-3\pi}{2}$ sq. units

(C) $\frac{28-5\pi}{2}$ sq. units

(D) $\frac{28-7\pi}{2}$ sq. units

Q.3

If $y = fx$ satisfies the condition $fx = f4 - x \quad \forall x \in (0, \quad 4), \quad fx = f14 - x \quad \forall x \in [4, \quad 10]$

$$fx = \begin{cases} x^2 - 2x + 5, & x \in (0, \quad 1] \\ 3 + \sqrt{2x - x^2}, & x \in (1, \quad 2] \\ \frac{20 - 2x}{3}, & x \in 4, \quad 7 \end{cases}$$

Then the area bounded by $y = fx$, $x = 0$, $x = 10$ and the x -axis is

(A) $\frac{\pi}{2} + 16$

(B) $\frac{\pi}{2} + 20$

(C) $\frac{\pi}{2} + 32$

(D) None of these

Q.4

Area between the curve $y = 2x^4 - x^2$, x axis and the ordinates of two minima of curve is

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(A) $\frac{7}{120}$

(B) $\frac{9}{120}$

(C) $\frac{11}{120}$

(D) $\frac{13}{120}$

Q.5

What is the area of the parabola $y^2 = 4bx$ bounded by its latus rectum ?

(A) $2b^2/3$ square unit

(B) $4b^2/3$ square unit

(C) b^2 square unit

(D) $8b^2/3$ square unit

Q.6

The area enclosed by the curve $x = a\cos^3 t, y = b\sin^3 t$ and the positive directions of x -axis and y -axis is

(A) $\frac{\pi ab}{4}$

(B) $\frac{\pi ab}{32}$

(C) $\frac{3\pi ab}{32}$

(D) $\frac{5\pi ab}{32}$

Q.7

The area bounded by the curve $y^2 = 4a^2(x-1)$ and the lines $\mathrm{x} = 1$ and $\mathrm{y} = 4\mathrm{a}$ is equal to :

(A) $\frac{16\mathrm{a}}{3}\mathrm{sq}$ unit

(B) $5\mathrm{a}$ sq unit

(C) $\frac{17\mathrm{a}}{4}$ sq unit

(D) None of these

Q.8

Let $fx = g1x^2 + xg^{'}x + g^{''}x$ and $gx = x^2 + xf^{'}1 + f^{''}2$, then area bounded by $y = gx$ and x-axis is

(A) 9

(B) $\frac{9}{2}$

(C) $\frac{27}{2}$

(D) 27

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Q.9

The area of the triangle formed by the tangent and normal at the point $(1, \sqrt{3})$ on the circle $x^2 + y^2 = 4$ and the x -axis is

(A) 3sq. units

(B) $2\sqrt{3}$ sq. units

(C) $3\sqrt{2}$ sq. units

(D) 4 sq. units

Q.10

Let $\mathrm{f}(\mathrm{x})$ be a continuous function such that the area bounded by the curve $\mathrm{y} = \mathrm{f}(\mathrm{x}), \mathrm{x}$ -axis and the lines $\mathrm{x} = 0$ and $\mathrm{x} = \mathrm{a}$ is $\frac{\mathrm{a}^2}{2} + \frac{\mathrm{a}}{2} \sin \mathrm{a} + \frac{\pi}{2} \cos \mathrm{a}$, then $\mathrm{f}\left(\frac{\pi}{2}\right)$ is

(A) 1.5

(B) 0.25

(C) 1.25

(D) 0.5

Q.11

The area of the region $\left\{(x, y) : 0 \leq y \leq x^2 + 1, 0 \leq y \leq x + 1, 0 \leq x \leq 2\right\}$ is

(A) 1.78

(B) 3.83

(C) 1.91

(D) 3.56

Q.12

The area bounded by the curve $y = \ln(x)$ and the lines $y = 0, y = \ln(3)$ and $x = 0$ is

(A) 1

(B) 2

(C) 3

(D) 4

Q.13

The line $y = mx$ bisects the area enclosed by lines $x = 0\, \mathrm{y} = 0$ and $\mathrm{x} = 3/2$ and the curve $\mathrm{y} = 1 + 4\mathrm{x} - \mathrm{x}^2$. Then the value of m is

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(A) 3.24

(B) 4.57

(C) 2.16

(D) none of these

Q.14

What is the area under the curve $y = |x| + |x - 1|$ between $x = 0$ and $x = 1$?

(A) $\frac{1}{2}$

(B) 1

(C) $\frac{3}{2}$

(D) 2

Q.15

Area bounded by the curves $y = \left[\frac{x^2}{64} + 2\right]$ ([.] denotes the greatest integer function), $v = x - 1$ and $x = 0$, above the x -axis is

(A) 2 sq unit

(B) 3 sq unit

(C) 4 sq unit

(D) None of these

Answers & Solutions

Q.1 Answer:

-3

Solution:

$f(x) = x^2 + bx - b; f'(x) = 2x + b \Rightarrow f'(1) = b + 2$

Equation of tangent:

$y - 1 = (b + 2)(x - 1)$ **Putting** $x = 0 \Rightarrow y = 1 - b - 2 = -b - 1 > 0 \Rightarrow b < -1$

Putting $y = 0 \Rightarrow x - 1 = -\frac{1}{b+2} \Rightarrow x = \frac{-1}{b+2} + 1$

$= \frac{b+1}{b+2} > 0 \Rightarrow b < -2$ **or** $b > -1$

Combining, the two conditions $= b < -2$

Now, $\frac{1}{2}|-b-1|\left|\frac{b+1}{b+2}\right| = 2; (b+1)^2 = 4|b+2| = -4b - 8$

$\Rightarrow (b+3)^2 = 0 \Rightarrow b = -3$ **follows the condition** $b < -2$

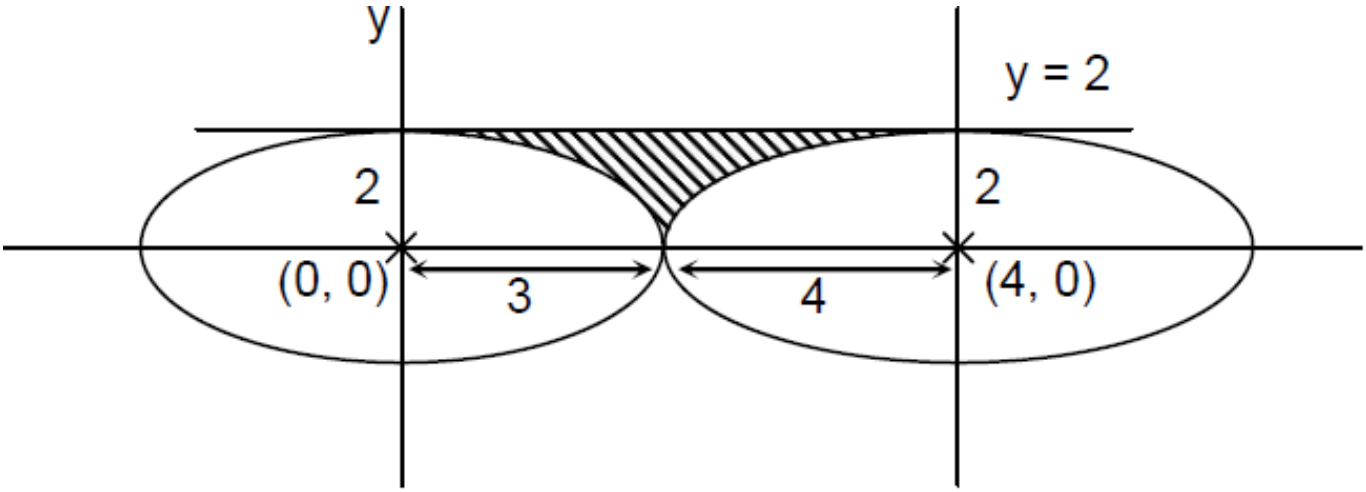
Q.2 Answer:

$\frac{28 - 7\pi}{2}$ sq. units

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Solution:

Area of an ellipse $= \pi ab$

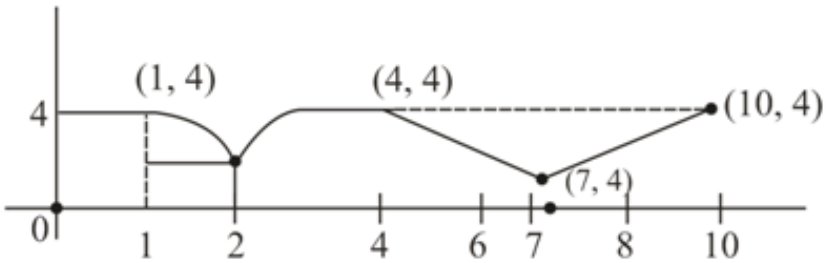


So, region $= 7 \times 2 - \frac{\pi}{4} \cdot 2 \times 3 - \frac{\pi}{4} \cdot 4 \cdot 2 = 14 - \frac{14\pi}{4} = 14 - \frac{7\pi}{2} = \frac{28 - 7\pi}{2}$

Q.3 Answer:

$\frac{\pi}{2} + 32$

Solution:



Required area $= 24 + \frac{\pi}{4} + 3 + \left(6 \times 4\right) - \frac{1}{2} \times 2 \times 6$

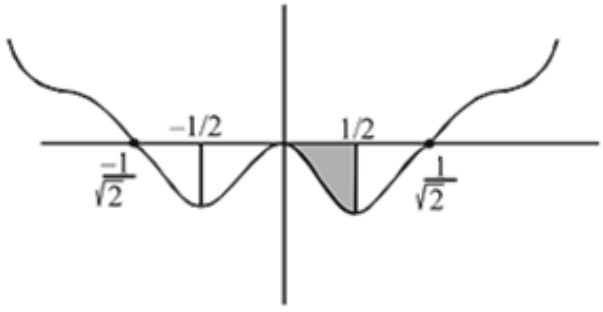
$= 32 + \frac{\pi}{2}$ sq. units

Q.4 Answer:

$\frac{7}{120}$

Solution:

The graph of $y = 2x^4 - x^2$ **is shown below :**



Minimum occurs at $x = \pm\frac{1}{2}$

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$\therefore$ **Desired area** $= 2\int_0^{1/2}(-y)dx = 2\int_0^{1/2}\left(x^2 - 2x^4\right)dx$

$= 2\left[\frac{x^3}{3} - \frac{2x^5}{5}\right]_0^{1/2} = 2\left[\frac{1}{24} - \frac{1}{80}\right] = \frac{7}{120}$

Q.5 Answer:

$8b^2/3$ square unit

Solution:

Given equation $y^2 = 4bx = 2\int_0^b \sqrt{4bx}dx$

$= 4\sqrt{b} \times \frac{2}{3}\left[x^{\frac{3}{2}}\right]_0^b = \frac{8\sqrt{b}}{3}\left[b^{\frac{3}{2}} - 0\right]$

$\therefore$ **area of parabola bounded by its latus rectum** $= \frac{8\,b^2}{3}$ **sq. units**

Q.6 Answer:

$\frac{3\pi ab}{32}$

Solution:

$y = 0$, **when** $t = 0$ **and then** $x = a$

So desire area

$A = \int_0^a y dx = \int_{\pi/2}^0 b\sin^3 t\left(-3a\cos^2 t\sin t dt\right)$

$A = 3ab\int_0^{\pi/2}\sin^4 t\cos^2 t dt = 3ab\int_0^{\pi/2}\cos^4 t\sin^2 t dt$

$\therefore 2\,\mathrm{A} = 3\mathrm{ab}\int_0^{\frac{\pi}{2}}\cos^2\mathrm{t}\sin^2\mathrm{t}\,\mathrm{dt} = 3\mathrm{ab}\cdot\frac{\pi}{16} \Rightarrow \mathrm{A} = \frac{3\pi ab}{32}$

Q.7 Answer:

$\frac{16a}{3}$sq unit

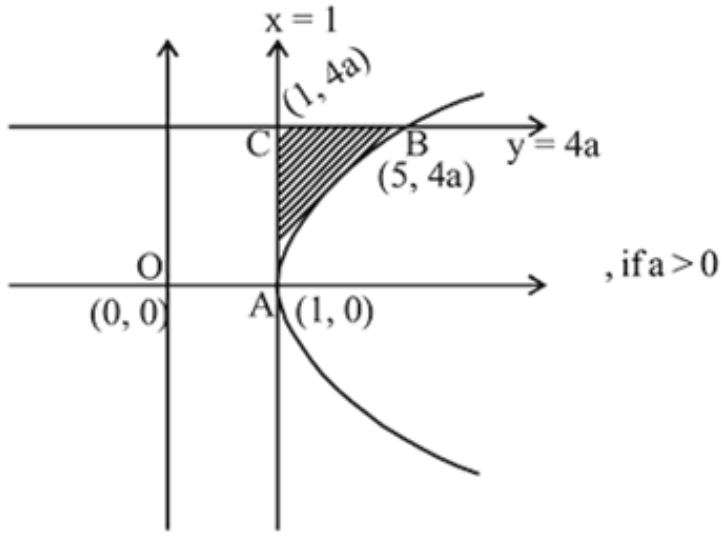
Solution:

Given curve : $y^2 = 4a^2(x-1)$

Lines : $\mathrm{x} = 1$ **and** $\mathrm{y} = 4\mathrm{a}$

Now, $(y-0)^2 = 4a^2(x-1)$

This is a parabola with vertex $\mathrm{A}(1,0)$.



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Required area = area of shaded region $\mathrm{ABC} = \int_1^5 [\mathrm{y}(\text{ line }) - \mathrm{y}\ (\text{parabola }\ )]\,\mathrm{dx}$
$= \int_1^5 [4\mathrm{a} - 2\mathrm{a}\sqrt{\mathrm{x}-1}]\mathrm{dx}$
$= \int_1^5 \left[4\mathrm{ax} - 2\mathrm{a} \times \frac{2}{3}(\mathrm{x}-1)^{3/2}\right]_1^5$
$= \left(20\mathrm{a} - \frac{32\mathrm{a}}{3}\right) - (4\mathrm{a} - 0) = \frac{16}{3}\mathrm{a}$ **sq.unit**

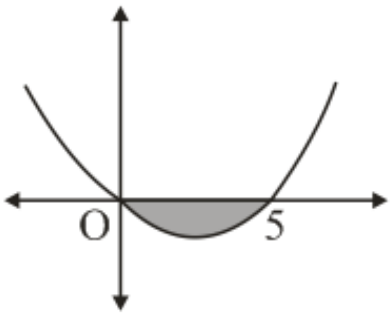
Q.8 Answer:

$\frac{9}{2}$

Solution:

Clearly, $fx = 2 - 3x$ **and** $gx = x^2 - 3x$

$gx =$



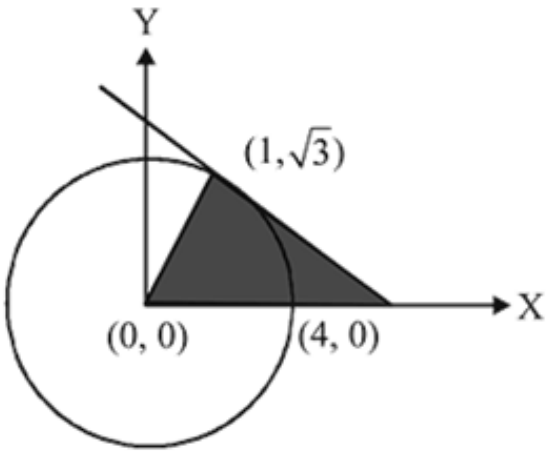
Area $= \int_0^3 x^2 - 3x\,dx = \frac{x^3}{3} - \frac{3x^2}{2}\Big|_0^3$
$= 9 - \frac{27}{2} = \frac{-9}{2} = \frac{9}{2}.$

Q.9 Answer:

$2\sqrt{3}$ sq. units

Solution:

The tangent on $x^2 + y^2 = 4$ **at** $(1, \sqrt{3})$ **is** $[\mathrm{x} + \sqrt{3}\mathrm{y} = 4]$
and equation of normal at $(1, \sqrt{3})$ **is** $y = x\sqrt{3}$



Required area $= \int_0^1 x\sqrt{3}\,\mathrm{d}x + \int_1^4 \frac{4-x}{\sqrt{3}}\,\mathrm{d}x$
$= \sqrt{3}\left[\frac{x^2}{2}\right]_0^1 + \frac{1}{\sqrt{3}}\left[4x - \frac{x^2}{2}\right]_1^4$
$= \sqrt{3} \times \frac{1}{2} + \frac{1}{\sqrt{3}}\left[4(4-1) - \frac{1}{2}(16-1)\right]$

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$= \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{3}}\left[12 - \frac{15}{2}\right] = \frac{\sqrt{3}}{2} + \frac{9}{2\sqrt{3}}$
$= \frac{\sqrt{3}}{2} + \frac{3\sqrt{3}}{2} = 2\sqrt{3}$ **sq. units**

Q.10 Answer:

0.5

Solution:

We have, $\int_0^a f(x)dx = \frac{a^2}{2} + \frac{a}{2}\sin a + \frac{\pi}{2}\cos a$

Differentiating w.r.t. a, we get

$\mathrm{f(a)} = \mathrm{a} + \frac{1}{2}(\sin \mathrm{a} + \mathrm{a}\cos \mathrm{a}) - \frac{\pi}{2}\sin \mathrm{a}$

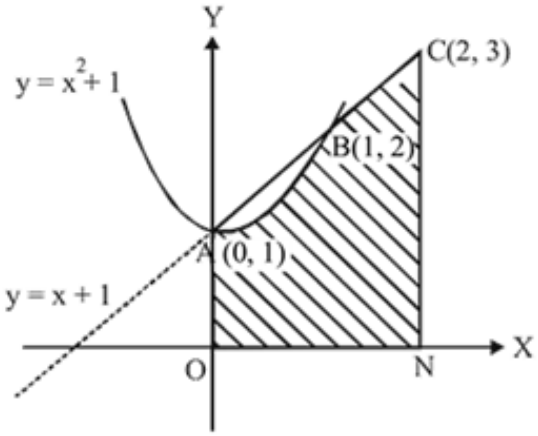
Put $\mathrm{a} = \frac{\pi}{2}; \mathrm{f}\left(\frac{\pi}{2}\right) = \frac{\pi}{2} + \frac{1}{2} - \frac{\pi}{2} = \frac{1}{2} = 0.5$

Q.11 Answer:

3.83

Solution:

The required region is the intersection of the following regions $\mathrm{A}_1 = \left\{(\mathrm{x}, \mathrm{y}) : 0 \leq \mathrm{y} \leq \mathrm{x}^2 + 1\right\}$ **. It represents the region below the parabola** $y = x^2 + 1$ $\mathrm{A}_2 = \{(\mathrm{x}, \mathrm{y}) : 0 \leq \mathrm{y} \leq \mathrm{x} + 1\}$**. It represents the region below the straight line** $y = x + 1$**, and** $\mathrm{A}_3 = \{(\mathrm{x}, \mathrm{y}) : 0 \leq \mathrm{x} \leq 2\}$**. It represents the region lying between the ordinates** $\mathrm{x} = 0$ **and** $\mathrm{x} = 2$**. The required area is the region shown as the shaded region in the following figure:**



By solving $\mathrm{y} = \mathrm{x} + 1$ **and** $\mathrm{y} = \mathrm{x}^2 + 1$**, we get the points of intersection** $\mathrm{A}(0, 1)$ **and** $\mathrm{B}(1, 2)$

Hence the required region is bounded by $y = f(x), y = 0 x = 0, x = 2$

Where $f(x) = \begin{cases} x^2 + 1, & 0 \leq x \leq 1 \\ x + 1, & 1 \leq x \leq 2 \end{cases}$

∴ Required area

$\mathrm{A} = \int_0^2 \mathrm{f(x)dx} = \int_0^1 \left(\mathrm{x}^2 + 1\right) \mathrm{dx} + \int_1^2 (\mathrm{x} + 1)\mathrm{dx}$

$= \left[\frac{x^3}{3} + x\right]_0^1 + \left[\frac{x^2}{2} + x\right]_1^2 = \frac{23}{6} = 3.83$

Q.12 Answer:

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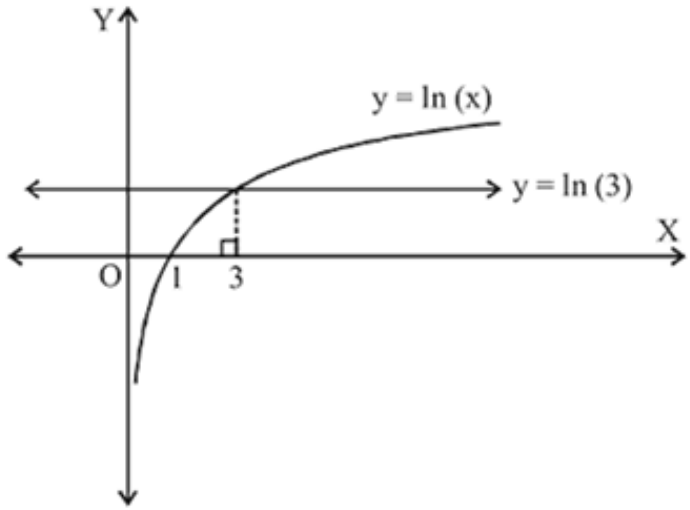
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Solution:

To find the point of intersection of curves $y = \ln(x)$ **and** $y = \ln(3)$**, put** $\ln(x) = \ln(3)$

$\Rightarrow \ln(x) - \ln(3) = 0 \Rightarrow \ln(x) - \ln(3) = \ln(1)$

$\Rightarrow \frac{x}{3} = 1 \Rightarrow x = 3$



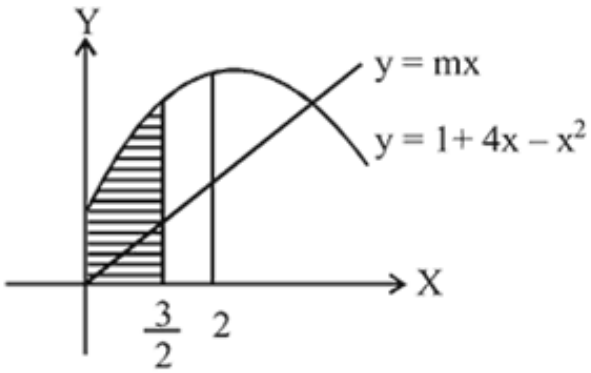
$$\text{Required area } = \int_0^3 \ln(3)dx - \int_1^3 \ln(x)dx$$
$$= [x\ln(3)]_0^3 - [x\ln(x) - x]_1^3 = 2$$

Q.13 Answer:

2.16

Solution:

$y = 1 + 4x - x^2 = 5 - (x-2)^2$



We have $\int_0^{3/2} \left(1 + 4x - x^2\right) dx = 2\int_0^{3/2} mxdx$

$= \frac{3}{2} + 2\left(\frac{9}{4}\right) - \frac{1}{3}\left(\frac{27}{8}\right) = \mathrm{m} \cdot \frac{9}{4}$

On solving we get $\mathrm{m} = \frac{13}{6} = 2.16$

Q.14 Answer:

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1

Solution:

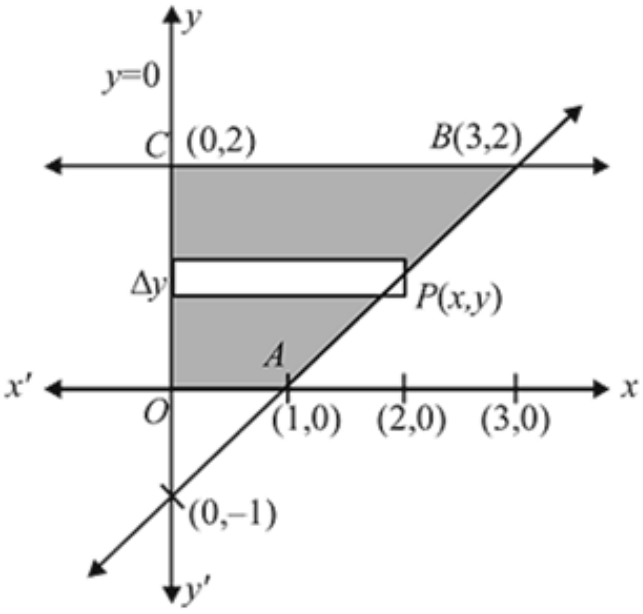
$|x|$ **for** $x \geq 0$
$= x$ **and** $|x-1|$ **for** $x \leq 1 = -(x-1)$
so, $\int_0^1 (|x| + |x-11|) =$ **required area**
$a = \int_0^1 x dx - \int_0^1 (x-1) dx$
$= \left[\frac{x^2}{2}\right]_0^1 - \left[\frac{x^2}{2} - x\right]_0^1 = \frac{1}{2} - \left(\frac{1}{2} - 1\right) = 1$ **sq units**

Q.15 Answer:

4 sq unit

Solution:

We have, $0 \leq \frac{x^2}{64} < 1$, **if** $-8 < x < 8$
$\Rightarrow 2 \leq \frac{x^2}{64} + 2 < 3$, **if** $|x| < 8$
$\Rightarrow y = \left[\frac{x^2}{64} + 2\right] = 2$, **if** $|x| < 8$
The graphs of the given curves is as shown in figure.



Req. area = area of the shaded region $= \int_0^2 x dy$
$= \int_0^2 (y+1) dy = \frac{1}{2}\left[(y+1)^2\right]_0^2 = \frac{9}{2} - \frac{1}{2} = 4$ **sq. unit.**

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